

POWER BI WEEK 6 REPORT

INTRODUCTION

Week 6 focuses on developing an interactive **Healthcare Analytics Dashboard** using the transformed healthcare data prepared in Week 5. Key performance indicators such as total patients, total billing amount, average billing amount, and average length of stay are created using DAX measures. Various visuals are used to analyse patient demographics, medical conditions, admission trends, hospital-related information, and billing patterns. Slicers are also added to enable dynamic filtering and cross-analysis of the dashboard.

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YEAR: II Year

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Figure 1: Total Patients measure created using COUNTROWS on the Billing table.

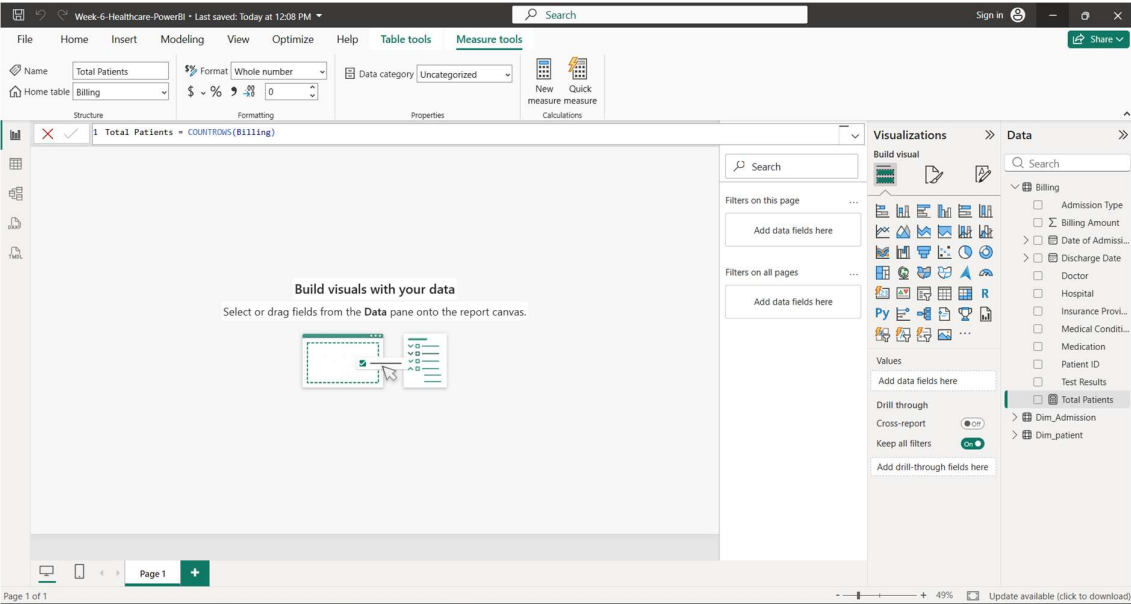


Figure 2: Total Healthcare Revenue measure created using the Billing Amount column.

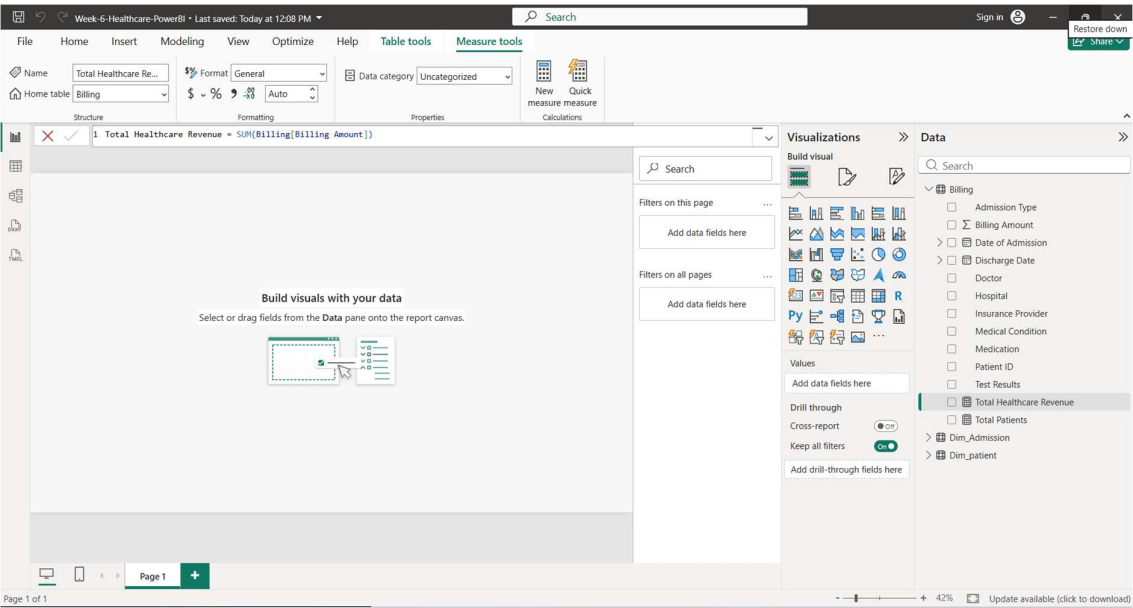


Figure 3: Average Billing Amount measure created using the Billing Amount column.

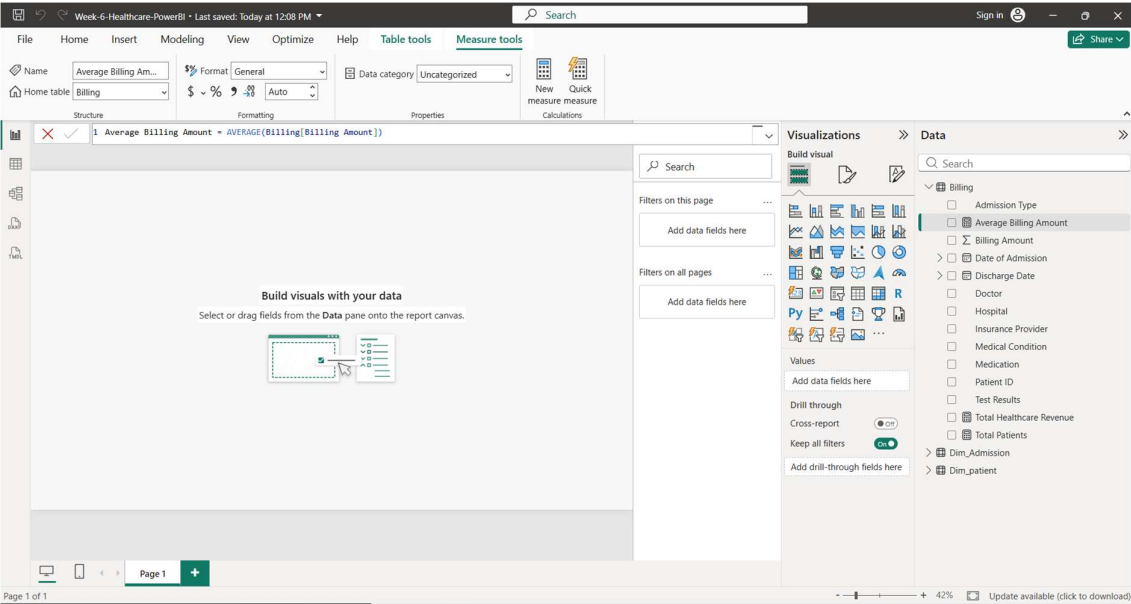


Figure 4: Average Length of Stay measure created using the admission and discharge dates.

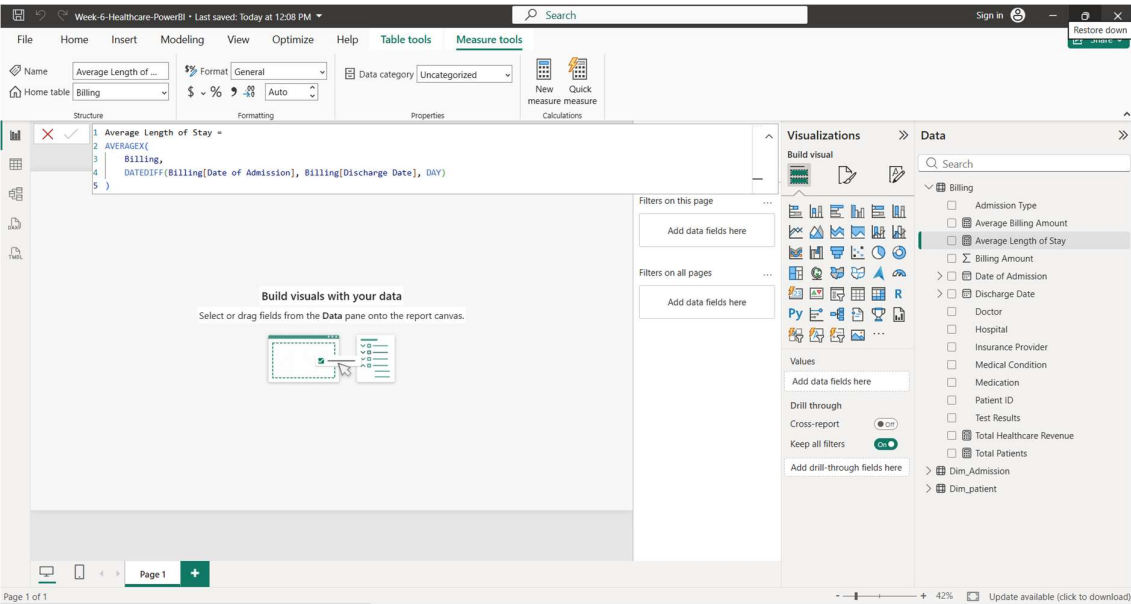


Figure 5: Total Billing Records measure created using COUNTROWS on the Billing table.

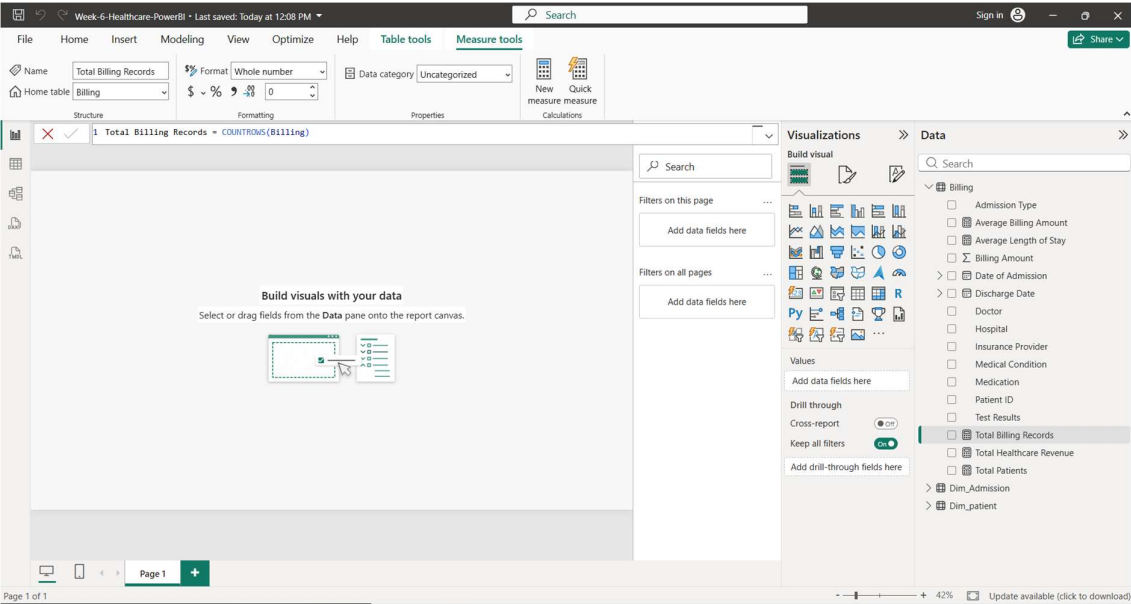


Figure 6: Total Patients KPI card displaying the total number of patients in the healthcare dataset.

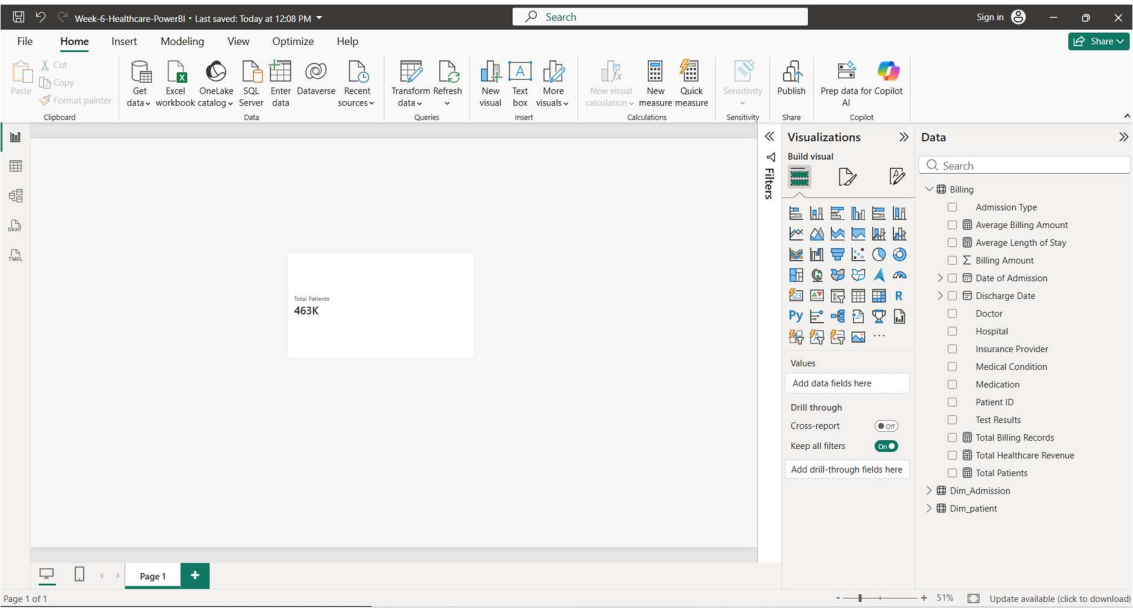


Figure 7: Total Healthcare Revenue KPI card displaying the total billing amount.

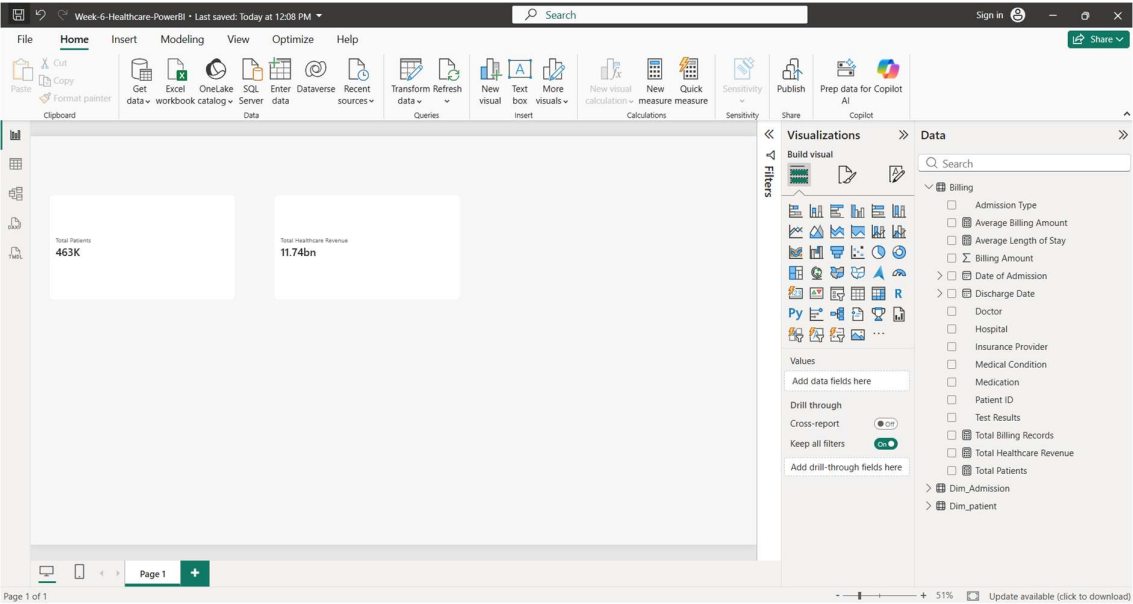


Figure 8: Average Billing Amount KPI card displaying the average billing amount per record.

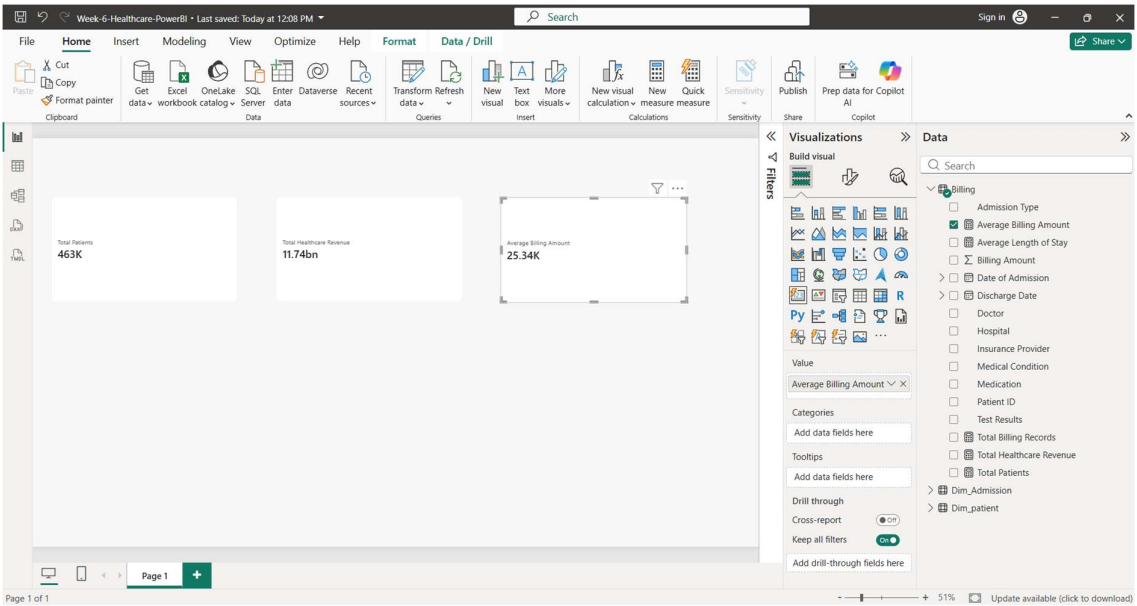


Figure 9: Average Length of Stay KPI card displaying the average number of days patients stayed in the hospital.

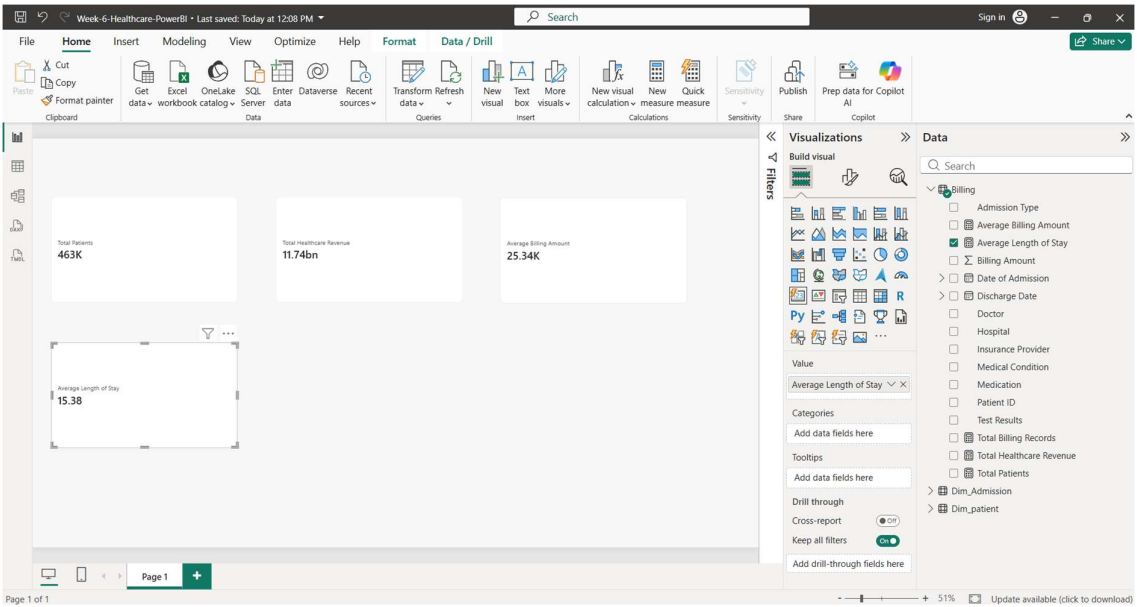


Figure 10: Total Billing Records KPI card displaying the total number of billing records.

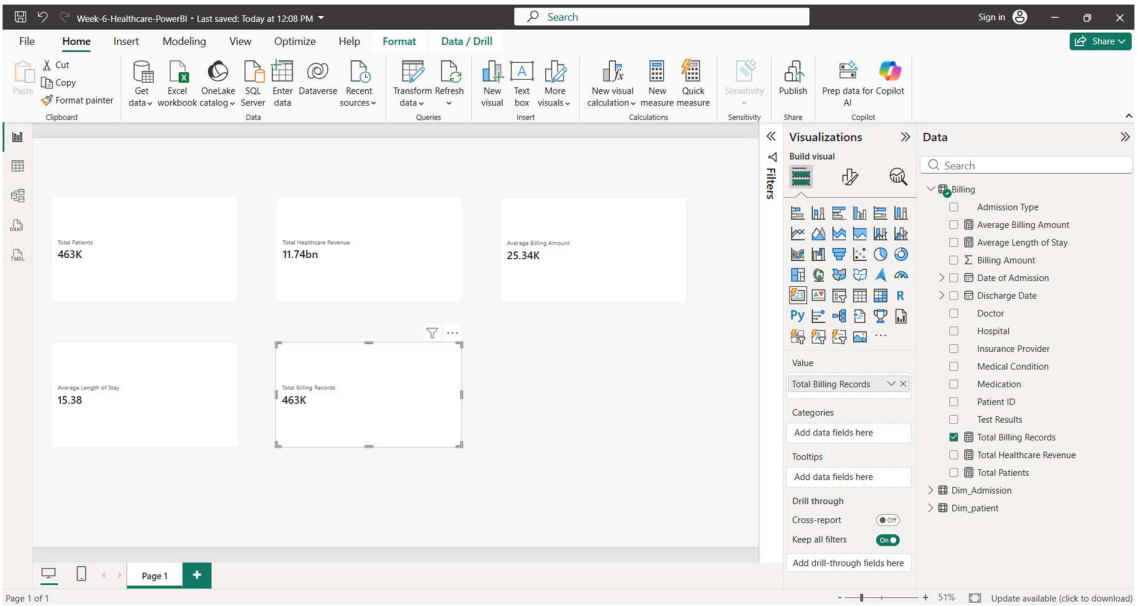


Figure 11: Column chart showing total billing amount by insurance provider.

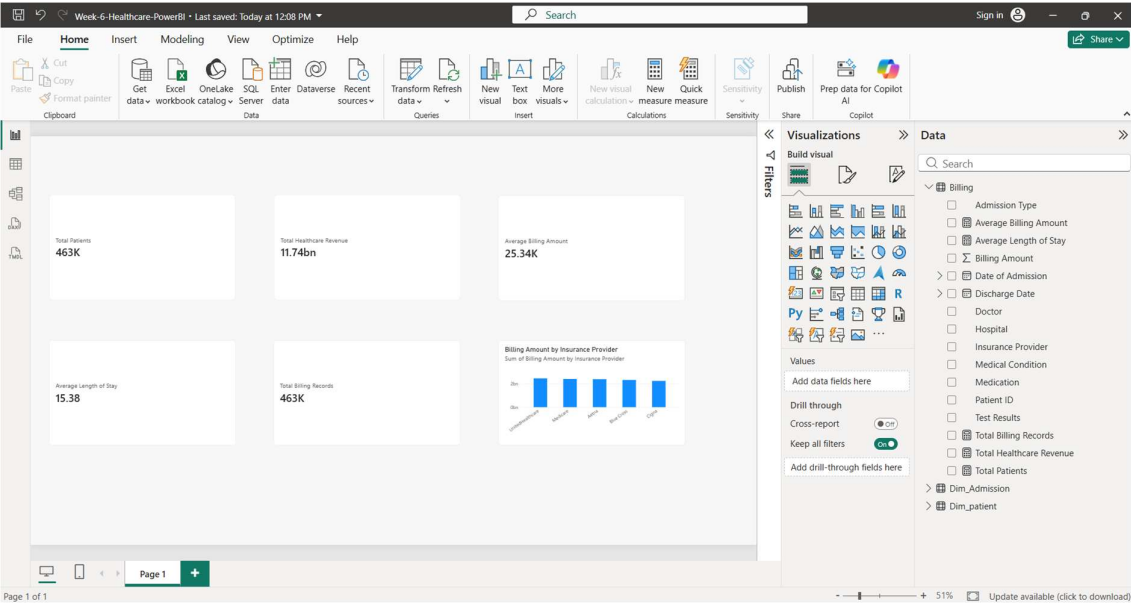


Figure 12: Donut chart showing patient distribution by gender.

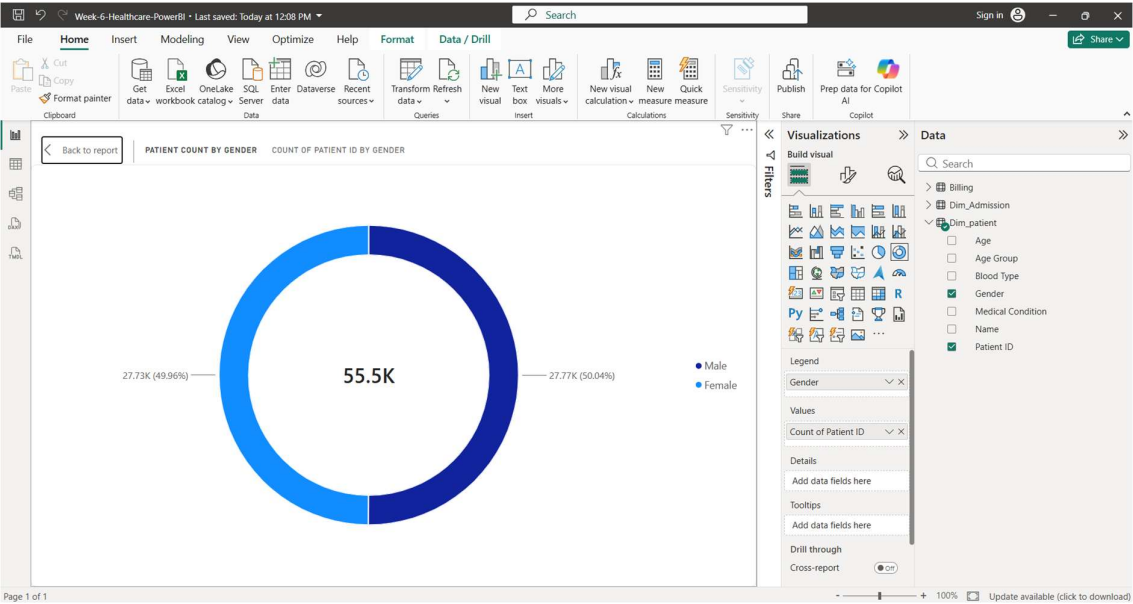


Figure 13: Bar chart showing total billing amount across different medical conditions.

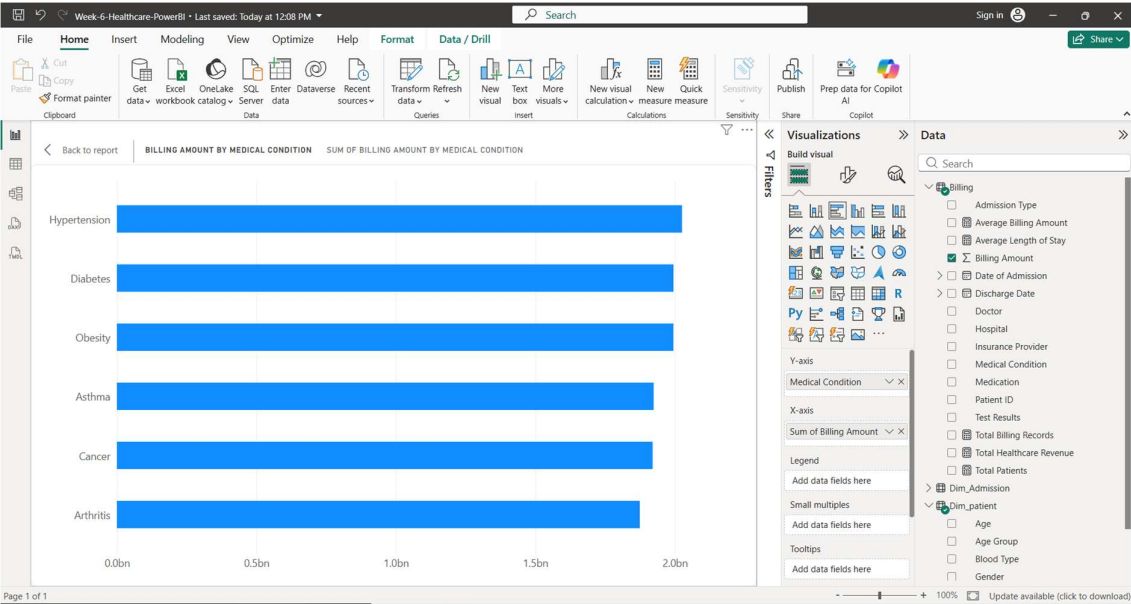


Figure 14: Column chart showing patient count across different age groups.

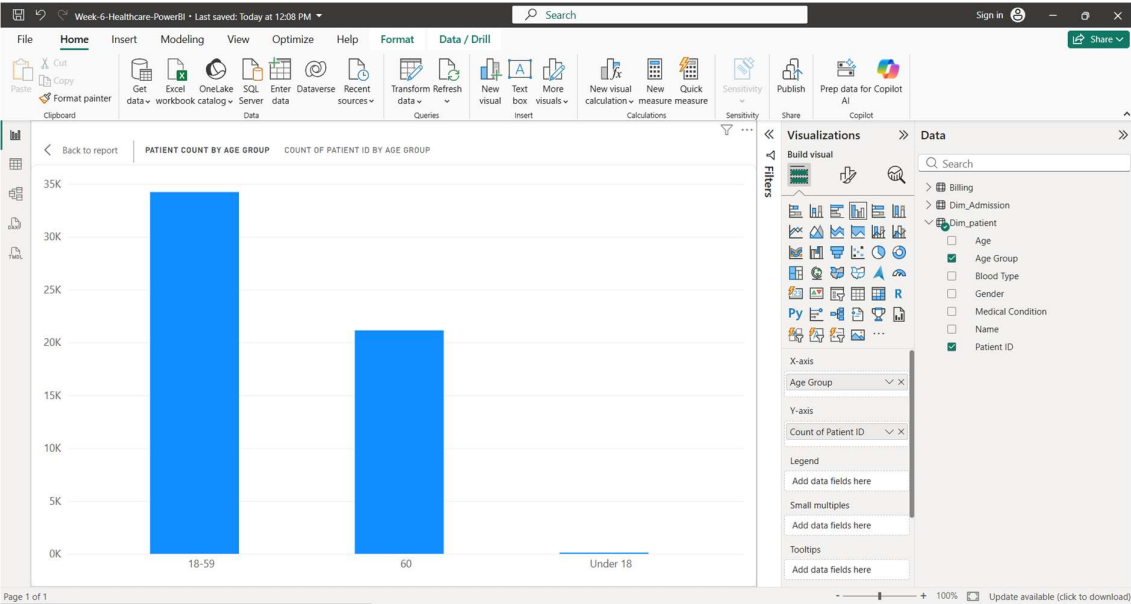


Figure 15: Column chart showing patient count across different medical conditions.

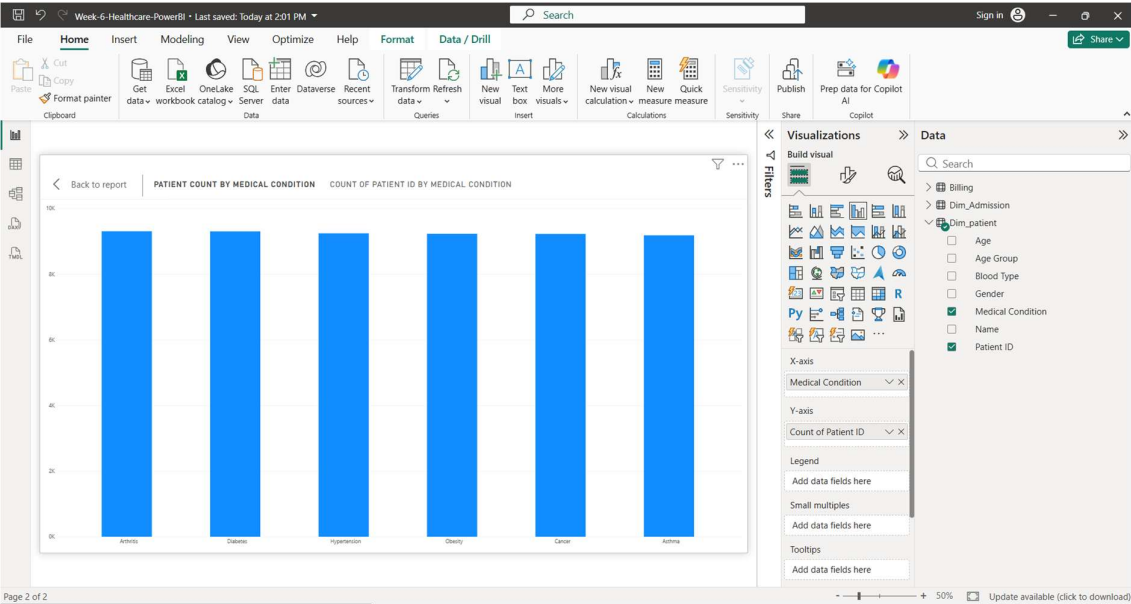


Figure 16: Matrix visual showing patient demographics by Age Group and Gender across different Medical Conditions, with Total Patients and Average Billing Amount.

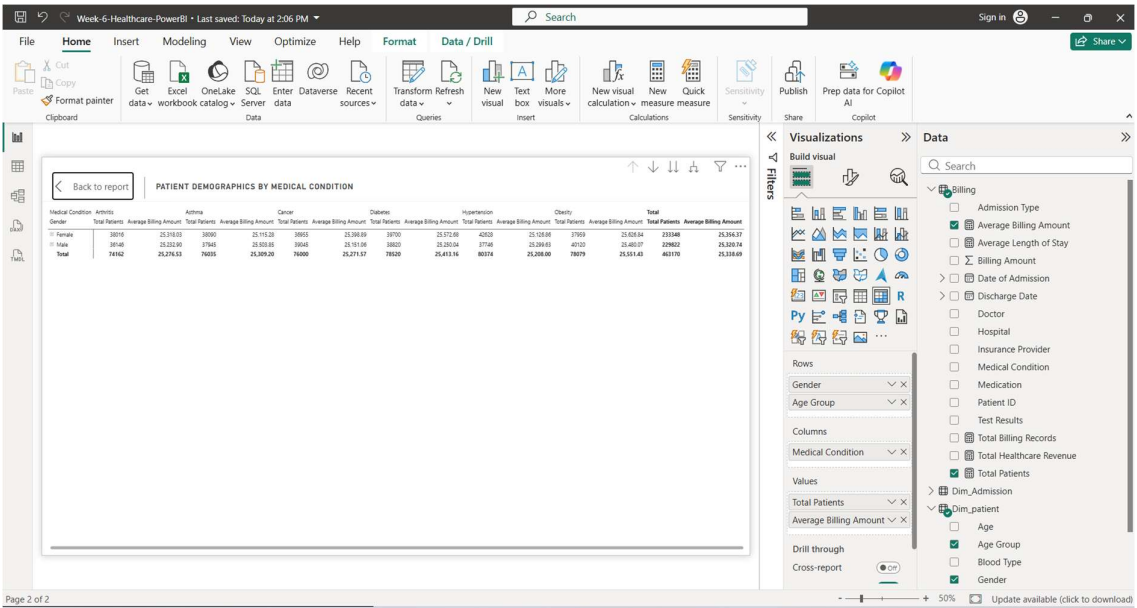


Figure 17: Donut chart showing patient distribution across different age groups.

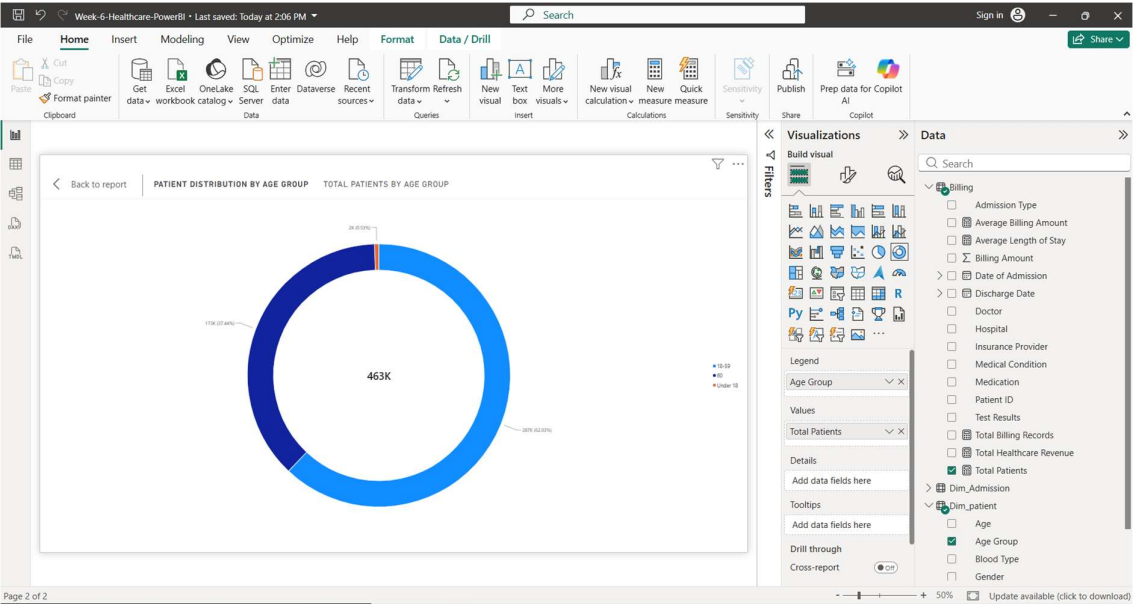


Figure 18: Stacked bar chart showing patient count by medical condition, split by admission type.

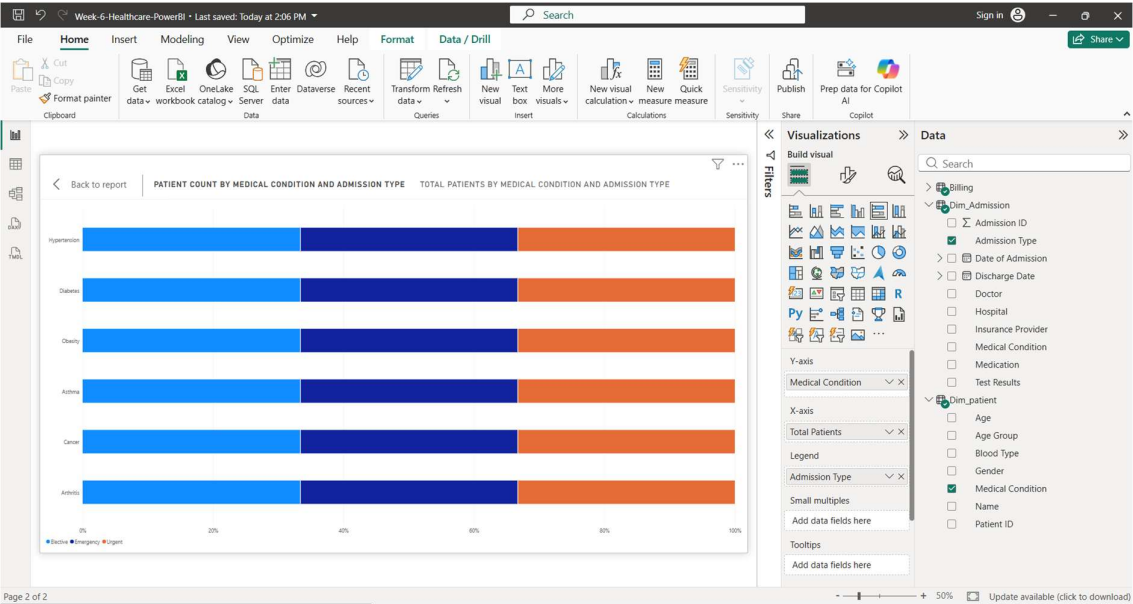


Figure 19: Line chart showing the trend in patient admission volume over time.

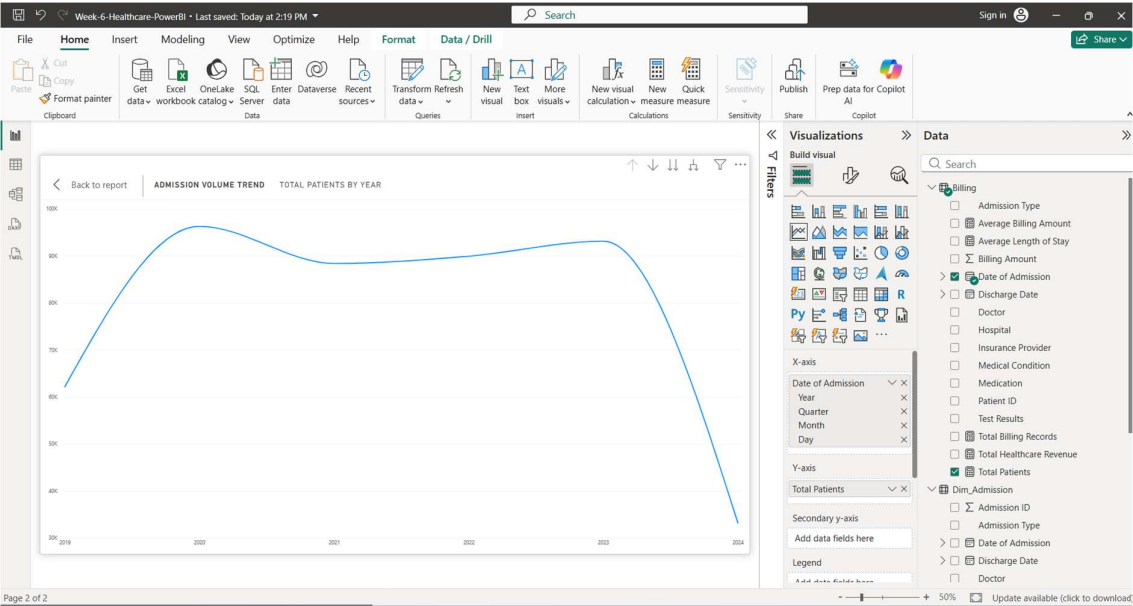


Figure 20: Clustered column chart showing total billing amount across different insurance providers.

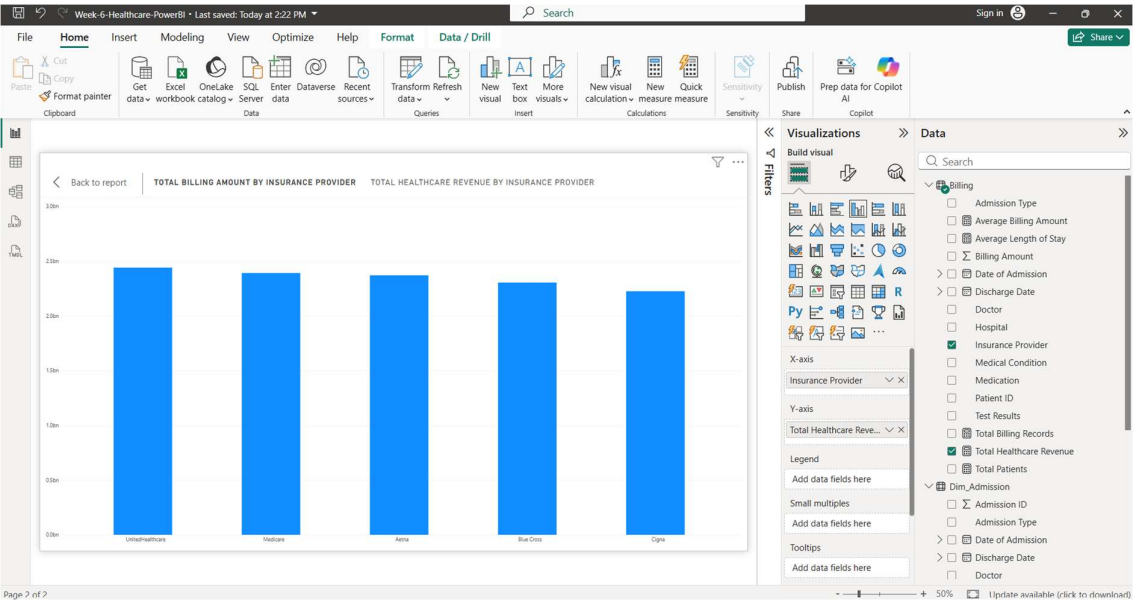


Figure 21: Readmission Rate measure created to identify patients with more than one admission.

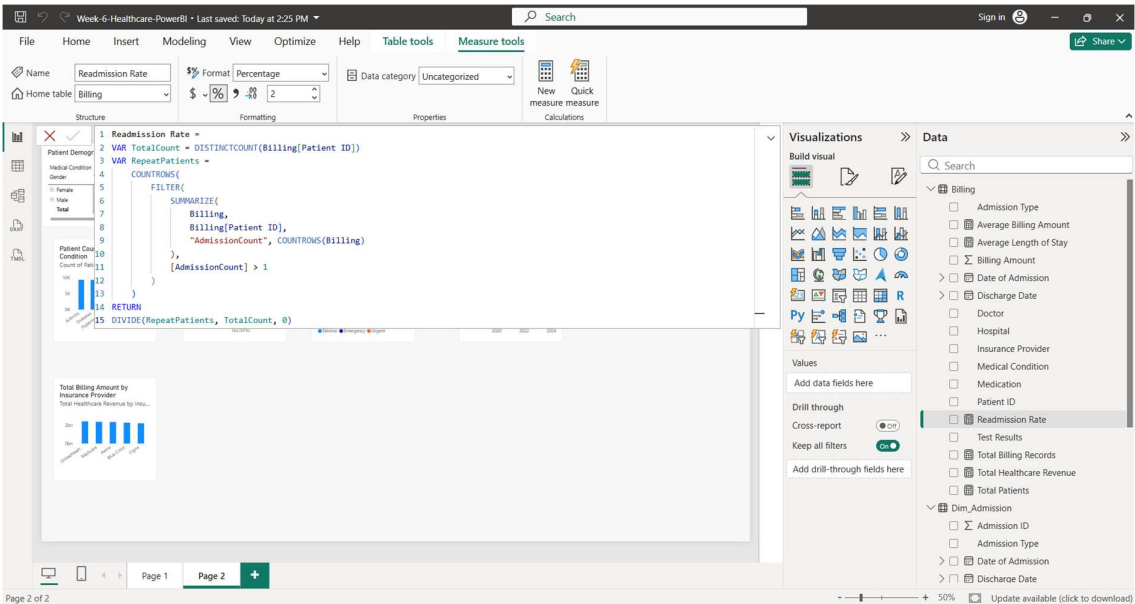


Figure 22: Insurance Provider slicer added for dynamic filtering of the healthcare dashboard.

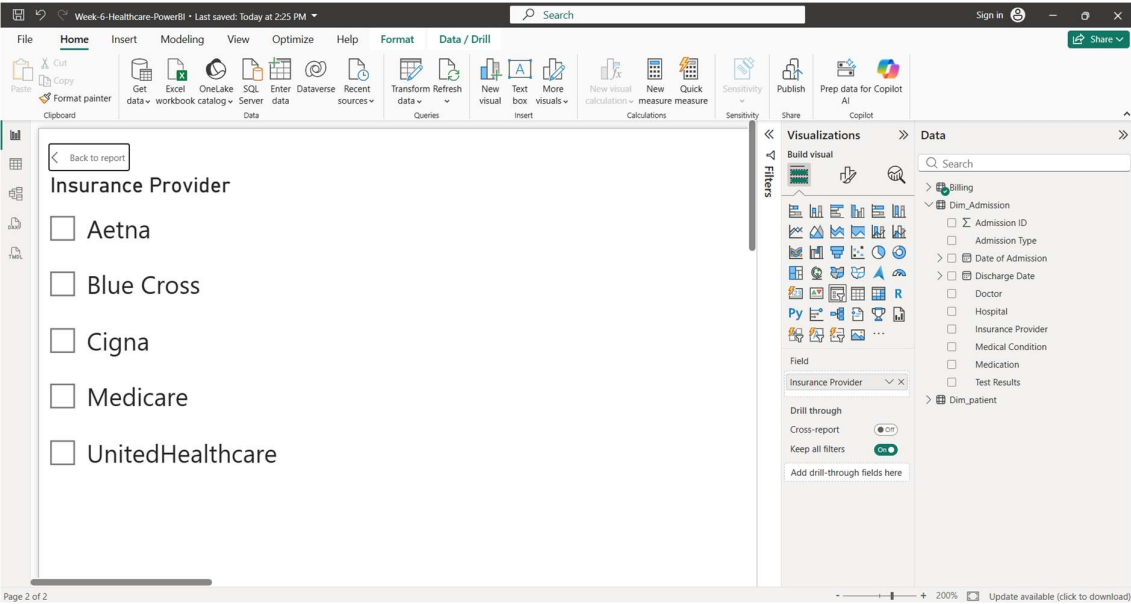


Figure 23: Admission Type slicer added for dynamic filtering of healthcare admission records.

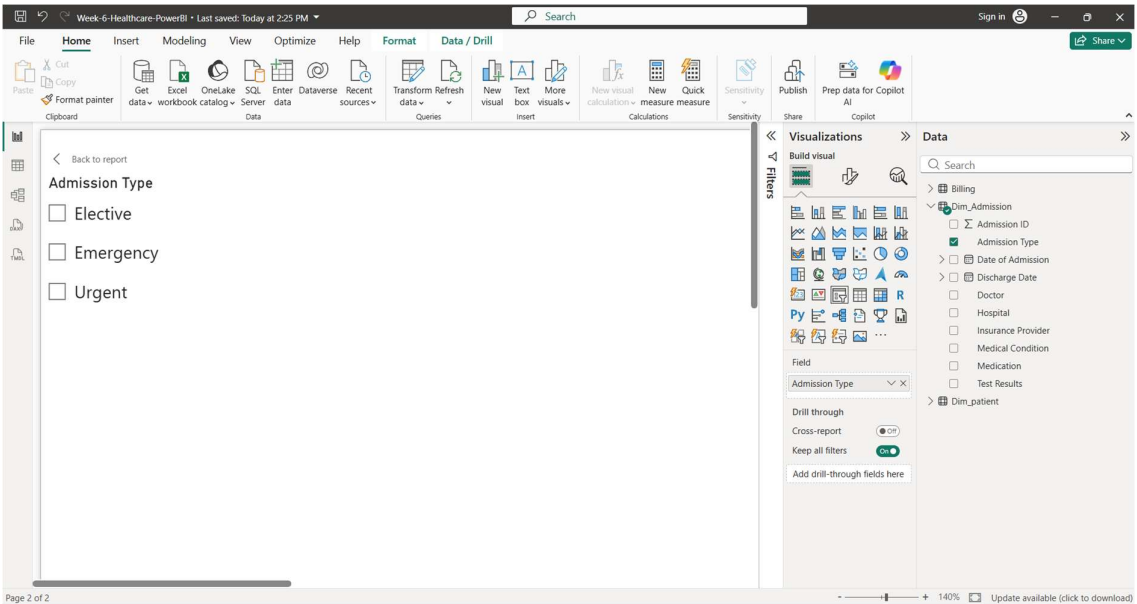


Figure 24: Date of Admission slicer added as a date-range filter for dynamic dashboard analysis.

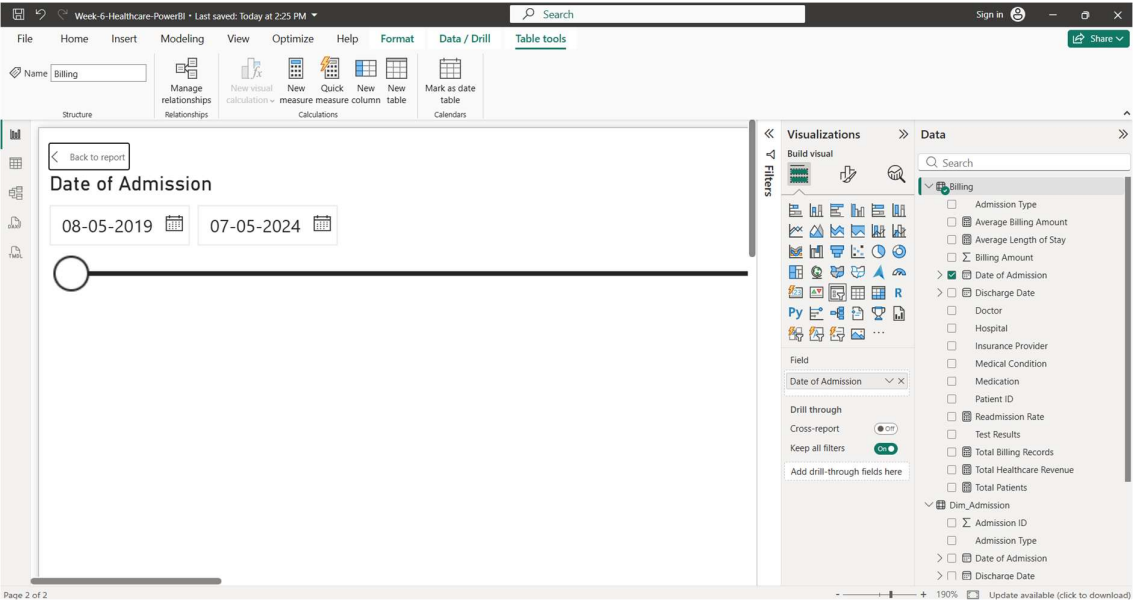
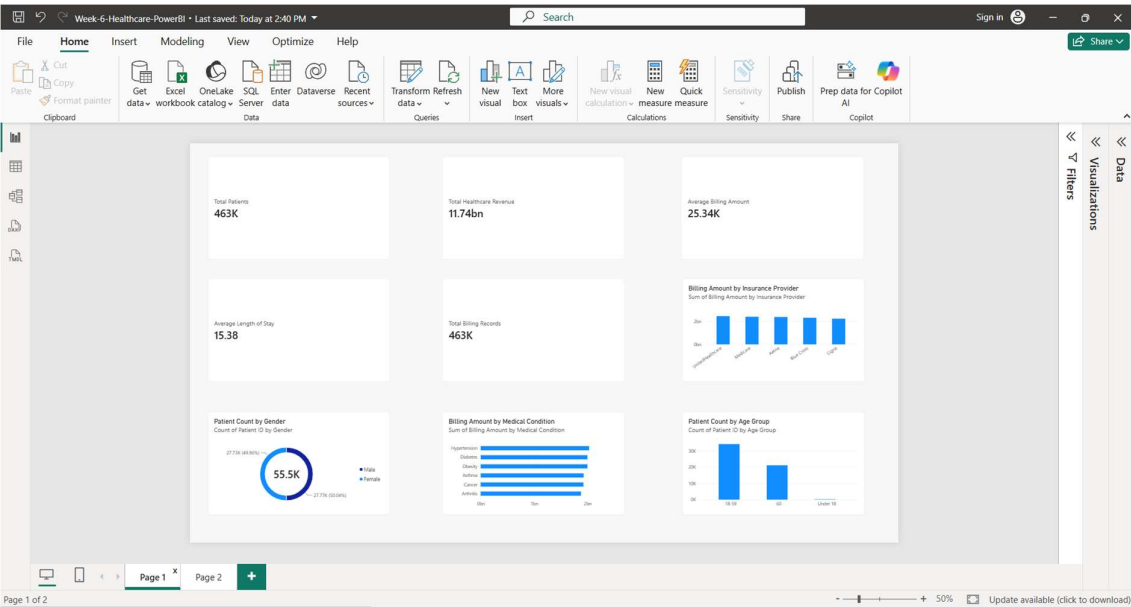
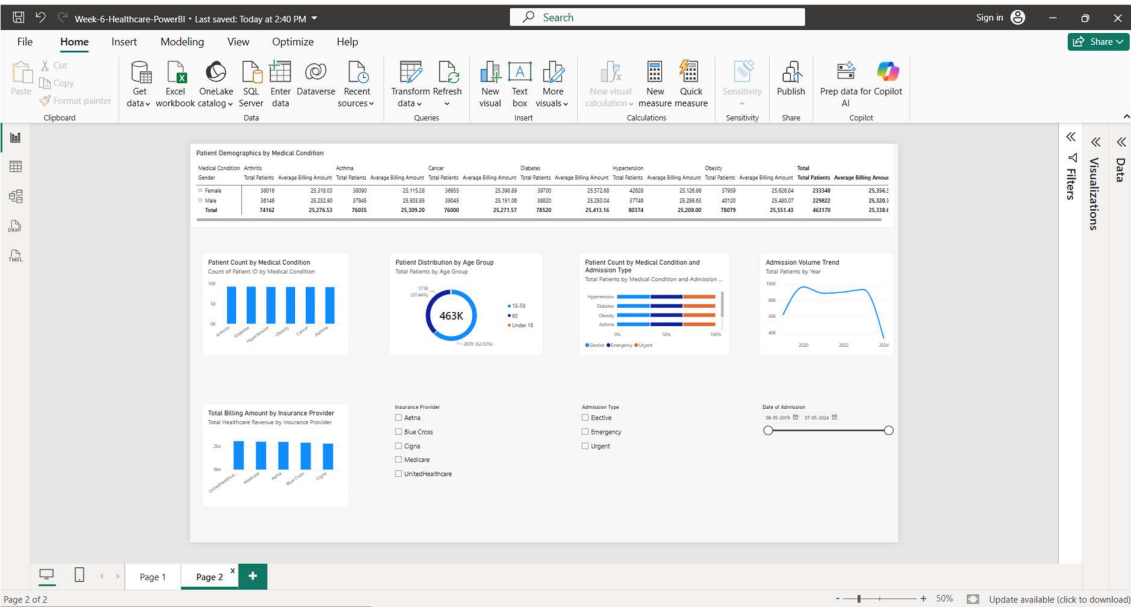


Figure 25: Final healthcare analytics dashboard integrating KPI cards, analytical visuals, and interactive slicers.

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Conclusion:

In Week 6, an interactive Healthcare Analytics Dashboard was developed using Power BI and DAX. Key measures such as Total Patients, Total Healthcare Revenue, Average Billing Amount, Average Length of Stay, Total Billing Records, and Readmission Rate were created. KPI cards, charts, a demographics matrix, and interactive slicers were added to analyse patient demographics, medical conditions, billing amounts, insurance providers, admission types, and admission trends. The final dashboard provides an interactive and structured view of the healthcare data for analysis and decision-making.